

U.S. Department
of Labor

Employment
and Training
Administration



0*NET Data Collection Program

Office of Management and Budget
Clearance Package Supporting
Statement and Data Collection
Instruments

December 2000

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A.1 Circumstances of Information Collection

This is a request for three-year OMB clearance to proceed with the *Occupational Information Network* (or *O*NET*) Data Collection Program. The *O*NET Data Collection Program* is planned to be a continuing activity to maintain a current database on the detailed characteristics of workers and occupations. The resulting O*NET database will be the most comprehensive standard source of occupational information in the United States. O*NET will be at the center of an extensive network of occupational information used by a wide range of audiences, including individuals making career decisions, public agencies and schools making training investment decisions, educational institutions preparing a future workforce, and employers making staffing and training decisions. O*NET also will provide a common language and framework to meet administrative needs of various federal programs, including workforce investment and training programs of the Departments of Labor, Education, and Health and Human Services. The uses of O*NET are described in further detail in **Section A.2**. Information about O*NET also can be found at the National O*NET Consortium's public website [<http://www.onetcenter.org>] and the U.S. Department of Labor, Employment and Training Administration's website, [<http://www.doleta.gov/programs/onet>].

A multiple method approach to updating O*NET has been developed. The primary method to update O*NET will be a survey of establishments and sampled workers from those establishments. In addition to the primary method, other methods will be used, including association member lists and subject matter experts. In all cases, the O*NET survey instruments will be used. More detailed information on the multiple methods can be found in **Section B.1.3**

A pretest was completed that provided insightful information on the impact of several proposed features on survey response rates, which in turn should maximize the return on investment during full-scale data collection. Specifically, the Pretest tested the use of different employee sample sizes per establishment and the use of a non-cash incentive for its impact on business response rates. It also tested alternatives on outreach, monetary incentives, and types of return envelope postage for their impact on employee response rates. A report documenting the pretest activity and results is provided in **Appendix E**.

The remainder of this section describes what O*NET is, O*NET's foundation, and its transition to the 1998 Standard Occupational Classification (SOC). This section also provides highlights of the pretest, an overview of the *O*NET Data Collection Program*, a summary of the data collection procedures, and a review of statutory and regulatory information.

A.1.1 What is O*NET?

O*NET is a comprehensive database system for collecting, organizing, describing, and disseminating information on occupational characteristics and worker attributes. O*NET replaces the Department of Labor's obsolete *Dictionary of Occupational Titles (DOT)* and provides additional information not available in the DOT. O*NET uses a data structure, the Content Model, to organize occupational information and provides a "common language" of standardized and defined descriptors and measures about occupations for use by all audiences. The O*NET Content Model is the result of extensive research and its development is fully documented ¹, as depicted in **Exhibit A-1**.

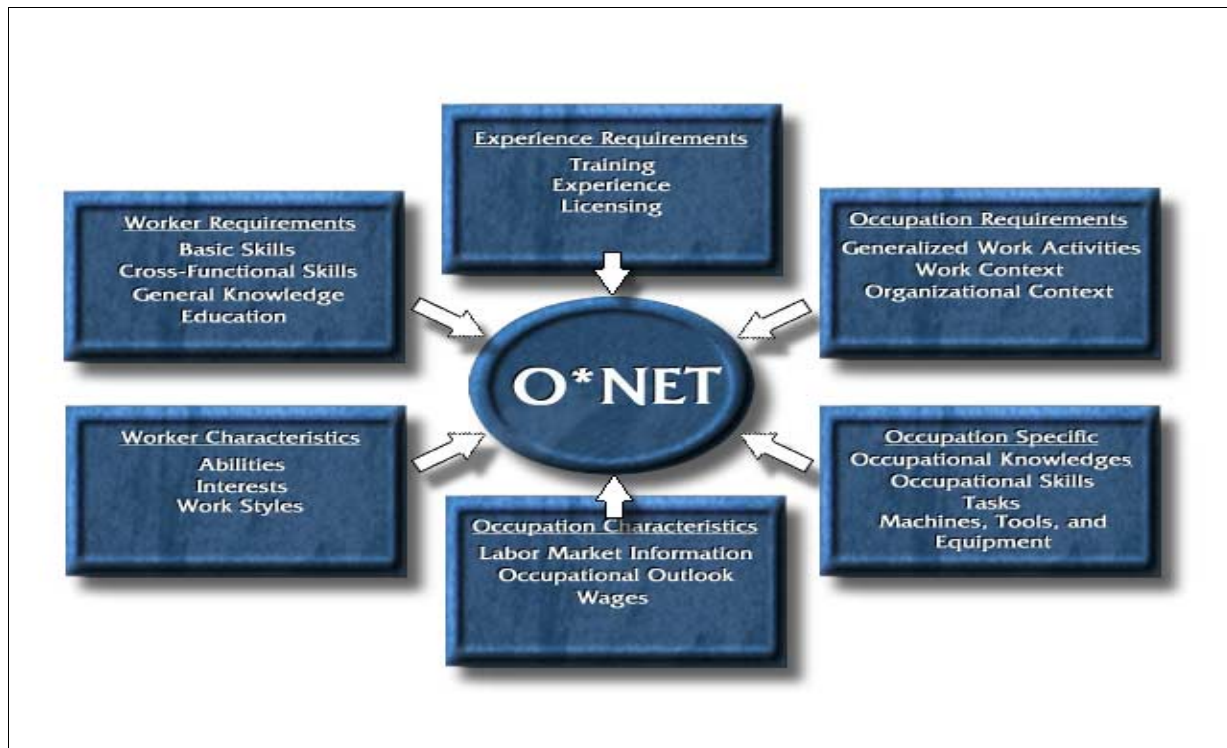
The Content Model is comprised of six domains:

1. *Worker Characteristics*: includes Abilities, Interests, and Work Styles
2. *Worker Requirements*: includes Basic Skills, Cross-Functional Skills, General Knowledge, and Education
3. *Experience Requirements*: includes Experience, Training, and Licensing Requirements
4. *Occupation Requirements*: includes Generalized Work Activities, Organizational Context, and Work Context
5. *Occupation Specific Requirements*: includes Occupational Skills and Knowledges, Tasks, Machines, Tools, and Equipment
6. *Occupation Characteristics*: includes information on industry, job outlook, and pay, based on existing data sources.

Each domain of the Content Model contains a hierarchical structure used to group specific detailed information. For example, the Worker Characteristics domain contains three types of information: *Abilities*, *Interests* and *Work Styles*. Abilities, in turn, contains four types of abilities: *Cognitive*, *Psychomotor*, *Physical* and *Sensory*. Each of these types of abilities contains further levels of detail. For example, Psychomotor abilities include *Fine Manipulative*, *Control Movement*, and *Reaction Time and Speed* abilities. Finally, the Fine Manipulative

¹ Utah Department of Employment Security, *Development of Prototype Occupational Information Network (O*NET) Content Model*. September, 1995; and *O*NET Final Technical Report*. September, 1996.

Exhibit A-1. O*NET Content Model



abilities contain three specific descriptors: *Arm-Hand Steadiness*, *Manual Dexterity*, and *Finger Dexterity*.

Examples of specific data in O*NET include mean rating data on level and importance for various questionnaire items (or descriptors), and text information on occupational definitions, descriptor definitions, scale anchors, and task descriptions. The descriptors and rating scales were developed through extensive research, primarily drawing from the field of job analysis in industrial/organizational psychology.² O*NET takes the best knowledge about both content and methodology from the last 60 years of research since the first DOT.

² *Development of Prototype Occupational Information Network (O*NET) Content Model*, Utah Department of Employment Security, September 1995. This document was authored by Norman G. Peterson and Michael D. Mumford of American Institutes for Research; Walter C. Borman, Personnel Decisions Research Institute, Inc.; P. Richard Jeanneret, Jeanneret & Associates, Inc.; and Edwin A. Fleishman, Management Research Institute, Inc., through a contract with the Utah Department of Employment Security and the American Institutes for Research.

Previous studies³ comparing different sources of job analysis ratings suggest that incumbents “...seemed best able to provide information across all descriptor domains. [In addition], large samples of knowledgeable job incumbents are available, which should contribute to the reliability of the resulting descriptive system.”⁴ Furthermore, the world of work is constantly changing and technological advancements are occurring so rapidly that an efficient and effective way to remain current and accurate is to obtain the information directly from the source — that is, the workers.

Workers selected as part of a random sample to participate in the *O*NET Data Collection Program* will be provided questionnaires and asked to rate the requirements of their own jobs as defined by the questionnaire items. The responses then will be tabulated into various statistics, such as mean ratings for each scale. Each item (or descriptor) in the O*NET questionnaires may use one or more scales. For example, the O*NET skill descriptor “Coordination” is rated both on a 5-point *Importance* scale, and a 7-point *Level* scale. Refer to **Appendix D** for the complete set of O*NET questionnaires.

The first version of the O*NET database released to the public was O*NET 98. O*NET 98 moved from the DOT’s over 12,000 occupations to a more user-friendly 1,122 “Occupational Units” (OUs), based on the Bureau of Labor Statistics’ Occupational Employment Statistics (OES) codes. The O*NET 98 Viewer, a stand-alone software application, accompanied the O*NET 98 database.

The current database version, O*NET 3.0, contains the same data used in O*NET 98; however, the occupations are restructured and coded to the 1998 Standard Occupational Classification (SOC). As a result of the restructuring and recoding, O*NET 3.0 has 974 occupations. O*NET 3.0 has a web-based accessing application called *O*NET OnLine*, which is available to the public at [<http://www.onetcenter.org>].

³ Peterson, N.G., Owens-Kurtz, C., Hoffman, R.G., Arabian, J.M., & Whetzel, D.C. (1990). Army synthetic validation project. Alexandria, VA: U.S. Army Research Institute for the Behavioral Sciences.; Fleishman, E.A., & Mumford, M.D. (1988). The ability requirement scales. In S. Gael (Ed.). *The Job Analysis Handbook for Business, Industry, and Government*. New York: Wiley.

⁴ Peterson, N.G., Mumford, M.D., Levin, K.Y., Green, J., & Waksberg, J. (1999, p. 34). Research method: Development and field testing of the content model. In Peterson, N.G., Mumford, M.D., Borman, W.C., Jeanneret, P.R., & Fleishman, E.A. (Eds.). *An occupational information system for the 21st century: The development of O*NET*. Washington, D.C.: APA.

O*NET 98 contained a total of 306 descriptors and 684 scales. The types of scales used for the O*NET ratings are *Importance*, *Level*, and *Frequency*. A review of O*NET 98-specific scales and descriptors during the preparation for pretest data collection led to some consolidation or dropping of descriptors and scales in order to reduce burden and increase the employee response rate.⁵ The next version of the O*NET 3.0 database will be restructured to incorporate improvements made to the O*NET data collection instruments. The improved instruments were used in the pretest and will be used again in the forthcoming data collection.

Exhibit A-2 summarizes the number of descriptors and scales included in the revised *O*NET Data Collection Program* questionnaires. We will collect information using 239 descriptors that include 400 scales contained in the revised Content Model domains. One of the revised domain questionnaires, Abilities, will be developed using trained analysts. No data collection is planned for the Occupational Characteristics domain. Information for Occupational Characteristics will be provided through linkages to the employment, wage, and long-term projections databases produced by the Bureau of Labor Statistics, the State employment security agencies, and other agencies.

The Foundation for O*NET

The *Dictionary of Occupational Titles (DOT)* was first published by the Department of Labor in 1939, and provided the occupational classification and descriptions needed for operation of the public Employment Service. The collection of information for the DOT was done primarily through observation and interview techniques conducted by trained occupational analysts. These techniques were applied to a small number of job incumbents from what can best be characterized as a “convenience” sample. The DOT has not been maintained or updated since 1991; many of its observations were conducted during the 1970s.

In the late 1980s, the Employment and Training Administration (ETA) of the U.S. Department of Labor began a review of the DOT program. The purpose was to address the concerns about the cost and difficulty of maintaining the DOT using the existing methods, and the need for additional and more current information. In 1990, as part of the DOT Review, the Secretary of Labor appointed the Advisory Panel on the DOT. In response to its charge to advise on improvements to the DOT, the Panel called for development of a new *database of occupational titles*, a concept which has evolved into O*NET. An important theme throughout

⁵See report, *Revisions to the O*NET Data Collection Instruments*, available at www.onetcenter.org.

Exhibit A-2. Summary of O*NET Data Collection Program Questionnaires

O*NET Data Collection Program Questionnaires	Number of Items and Scales			Data Source
	Number of Items	Number of Scales for Each Item	Total Number of Scales	
Skills	35	2	70	Incumbents ^a & Analysts
Knowledge	33	2	66	Incumbents ^a
Work Styles ^b	16	1	16	Incumbents ^a
Education & Training ^b	5	1	5	Incumbents ^a
Generalized Work Activities	41	2	82	Incumbents ^a
Work Context	57	1	57	Incumbents ^a
Abilities	52	2	104	Analysts
Tasks ^c	varies	2	varies	Incumbents ^a
Total (not including tasks)	239		400	

^a Subject matter experts will rate the same questionnaires as incumbents for those occupations that will use this method.

^b The Knowledge Questionnaire packet also includes the Work Styles and Education & Training Questionnaires.

^c All incumbents will be asked to complete a task questionnaire in addition to the domain questionnaire they are asked to complete.

the Panel's recommendations was the development of a "common language" about jobs, occupations, and skills. **The Panel's complete report is presented in *Appendix A*.**

Transitioning O*NET to the Standard Occupational Classification

The Office of Management and Budget has recently instructed all government agencies to implement the 1998 Standard Occupational Classification (SOC). This transition has been successfully completed for O*NET, which is now closely compatible with the SOC. The number of occupations has been reduced from 1,122 in O*NET 98 to 974 in O*NET 3.0. In most cases, an O*NET occupation is equivalent to a single SOC occupation. In some cases, the O*NET occupations provide more detail within a single SOC occupation.

A.1.2 O*NET Data Collection Approach

The *O*NET Data Collection Program* is the next critical step in the full development of O*NET. In the research leading to the *O*NET Data Collection Program*, various information development methods were examined, including collection of data from job incumbents and supervisors, and development of ratings by subject matter experts and occupational analysts. Based on this work, the O*NET Team has determined that the preferred source of data for most domains is job incumbents, while analysts are preferred for the Abilities domain, which tends to be more abstract. In addition, other subject matter experts such as supervisors and trainers may be used where access to job incumbents may be problematic.

In addition, the Advisory Panel on the DOT recommended the use of sampling techniques to ensure the representativeness and accuracy of the occupational data. By contrast, no systematic sampling methods were used in the development of the original Dictionary of Occupational Titles.

Collecting representative information from job incumbents presents many challenges; among them is determining the best method for identifying a representative sample of job incumbents in each occupation. This problem must be solved regardless of whether the data are gathered through self-administered surveys, personal interviews, or analyst observation. In particular, O*NET is concerned with identifying sampling approaches that minimize burden on employers and the public, achieve broad coverage of the workers in each occupation, allow acceptable response rates to be achieved, and are cost-effective.

Three types of sampling frames are available for identifying samples of workers in each occupation: households, employers, and lists of individual workers identified through professional and trade associations, licensing agencies, and unions. Each of these options has been examined in light of the criteria mentioned above (burden, coverage, response rates, and cost-effectiveness), and each offers advantages and disadvantages.

A household survey would provide the advantage of broad coverage of employment, as all types of workers – wage and salary, self-employed, and unpaid family workers – would be surveyed. Potential response rates for an O*NET household survey have not been tested, but might be inferred from other self-administered mail household surveys. For example, the Census 2000 long form is a self-administered questionnaire similar in response time required

(38 minutes⁶ compared to about 30 minutes for O*NET), although the questions are different in nature. The Census 2000 response rate for the long form was 54.1 percent.⁷

The household survey approach has other significant disadvantages for O*NET, primarily related to burden and cost-effectiveness. O*NET requires a representative sample of workers in each of over 900 specific occupations. Employment in many of these occupations is small – under 100,000 nationwide for over half the occupations⁸ – indicating that a very large and costly household sample would be required to find an adequate sample of workers for all of the O*NET occupations. Unlike with employer surveys, it would be very difficult to target the sample to find households that are most likely to have workers in specific occupations.

The employer survey approach, as proposed in this clearance request, provides the advantage of lower response burden and cost when compared with a household survey, as well as good coverage for the large majority of occupations. Response burden and cost are lower for two primary reasons. First, there are more workers per employer than per household. Thus, the number of contacts required to identify workers is fewer. Second, employer contacts can be minimized by focusing the survey on employers who are most likely to employ workers in each specific occupation for which the sample is needed. Efficient sample design is possible because the distribution of employment in an occupation is usually a function of the industry of the employer. Employment by occupation by industry is measured by the Occupational Employment Statistics program of the Bureau of Labor Statistics.

The employer survey provides good coverage of wage and salary employment as long as an acceptable employer sampling frame is available. Coverage of self-employment is more difficult, although the sampling frame planned for use in O*NET surveys includes many businesses operated by self-employed workers. Coverage of unpaid family workers in an employer sampling frame is minimal or zero, but the number of workers involved is quite small (135,000 nationwide in 1999⁹).

⁶www.census.gov, Census 2000 Frequently Asked Questions.

⁷Bureau of the Census, Economic and Statistics Administration, response to public information e-mail inquiry, October 25, 2000.

⁸Employment data on the 1998 Standard Occupational Classification are not yet available. Using the Bureau of Labor Statistics Occupational Employment Statistics classification, however, 1998 employment data shows that employment was less than 100,000 in 542 of the 777 occupations for which estimates were produced. Employment was less than 50,000 for 421 occupations, and less than 10,000 for 141 occupations.

⁹Bureau of Labor Statistics, Employment and Earnings, January 2000.

Achieving high response rates with the employer survey approach can be problematic, as this approach requires cooperation at two levels—first by the employer and then by the sampled worker. Experience with the O*NET pretest documents the response rates achieved using pretest methods and incentives, where the “best design” produced a 69 percent employer cooperation rate and a 55 percent employee response rate. Plans to improve response rates are discussed in **Section B.3**.

Identifying sampling frames of workers through professional and trade associations and unions has also been identified by the O*NET Team as an approach for some occupations. The advantages of this method are reduction in response burden, as contacts with a sample of employers is replaced with contact with one or a few associations. This procedure also removes the cost of soliciting and maintaining employer cooperation, but adds the cost of soliciting and maintaining association cooperation. Response rates among association members may prove to be higher than found among workers in samples provided by employers, as the association involvement may encourage response. The calculated response rates will be higher, as the problem of employers not providing the questionnaires to workers will be eliminated (see discussion of this problem in **Section B.1.2**).

A major disadvantage of the association member sample is coverage. This method can be used only where association membership encompasses a significant share of employment in the occupation, which may be difficult to identify accurately. Where coverage of employment in specific occupations by association membership is significant but not broad enough, the association approach can be used along with an employer sample in a dual-frame sample design, as described in **Section B.1.4**.

In summary, the potential sampling frames for the O*NET survey – households, employers, and association memberships – each offer advantages and disadvantages when evaluated according to burden, response rates, coverage, and cost-effectiveness. The O*NET Team has concluded that the employer sampling frame approach provides the best method for most occupations, and the association member sample should be used for a small number of occupations, either in a dual-frame or alone, depending on the coverage provided by the association list. With both approaches, continuous improvement in survey design and methods are planned to increase response rates.

A.1.3 Pretest Highlights

In the pretest, information was collected in a two-stage design. Stage 1 included a statistical sample of businesses expected to employ workers in the specific occupations being surveyed. Stage 2 included a sample of workers in the selected occupations within the sampled businesses. These workers were asked to complete the survey instruments.

Presented below are various pretest experiments and their respective results from the *O*NET Data Collection Pretest*. First, specific pretest methodological experiments are described. Second, business eligibility and employee response rates are highlighted. Finally, results of the methodological experiments are presented.

Pretest Methodological Experiments

The pretest included the following experiments to assess their impact on response rates:

- # Non-cash incentive (Work Analysis Toolkit) for businesses
- # Type of return envelope postage (business reply vs. first class stamps) for employees
- # Inclusion of a short informative video on O*NET in the questionnaire packets for employees
- # Seven cash incentive plans for employees, including no incentive, post-incentives only, pre-incentives only, and “hybrid” pre- and post-incentives. ***Exhibit A-3*** presents the different plans.

Business and Employee Response Rate Results

- # 6,881 businesses were sampled
- # 6,399 businesses were screened (93%)
- # 2,055 business were eligible (32%)
- # 1,149 businesses were recruited
 - ▶ business recruitment rate was 68% using the Toolkit incentive
 - ▶ business recruitment rate was 52% without the Toolkit incentive

Exhibit A-3. Pretest Cash Incentive Plans

Incentive Plan		Pre-Response Incentive Amount	Post-Response Incentive Amount	Total Incentive Amount
1	No Incentive	\$0	\$0	\$0
2	Pre-Response Incentive Only	\$5	\$0	\$5
3	Pre-Response Incentive Only	\$10	\$0	\$10
4	Post-Response Incentive Only	\$0	\$10	\$10
5	Post-Response Incentive Only	\$0	\$20	\$20
6	Both Pre- and Post-Response Incentives	\$5	\$10	\$15
7	Both Pre- and Post-Response Incentives	\$10	\$10	\$20

- # 3,766 employees were sampled
- # 1,662 employees responded, with response rates varying by experiment
 - highest employee response rate was 55%
 - lowest employee response rate was 31%
- # No difference in raw response rates were detected by type of questionnaire employee completed
- # There was no significant item non-response on the individual questions. Item completion rates ranged from 92% to 100%, averaging 99%
- # 3.7% of responses were received via the Internet, providing a threshold for improving the use of the Internet in future surveys.

Results of Methodological Experiments

Business Incentive

- # A Work Analysis Toolkit was provided to 25% of the eligible businesses.
- # The recruiting rate was higher for businesses offered the Toolkit (68% vs. 52%, $p = 0.001$).

Video Tape Inclusion

- # The video had no effect on employee response rates ($p = 0.5$).

Return Envelope Postage

- # A two-percentage point increase in employee response rates was experienced by the first class stamped return envelopes over the business reply envelopes; however, this difference was not statistically significant ($p = 0.4$).

Cash Incentives

- # Results of this experiment revealed that any cash incentive was more effective than none.
- # The pre-incentive was more effective than the post-incentive.
- # The hybrid incentives were not significantly different from pre-incentive only conditions.

Overall “Best Design”

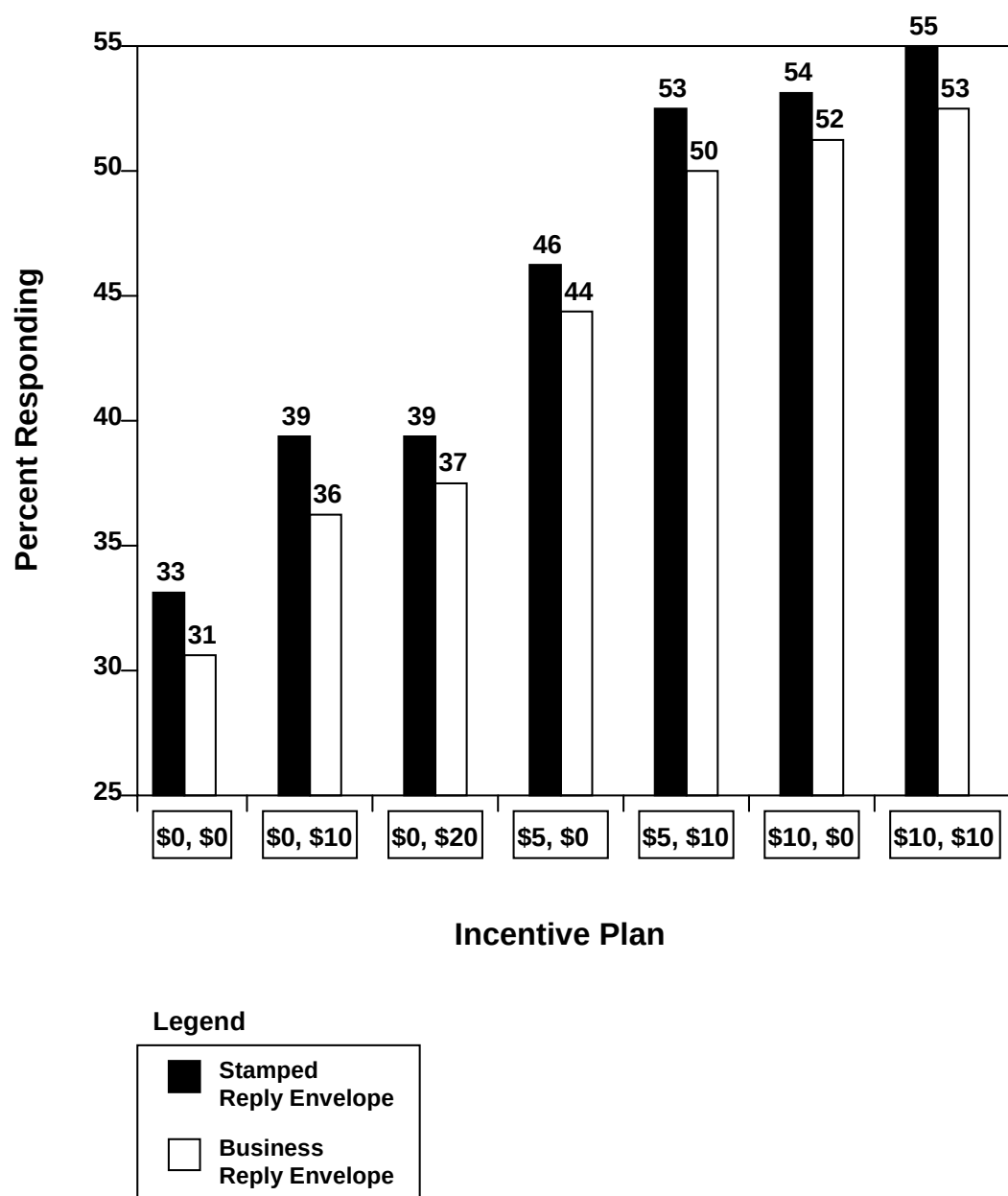
- # Include non-cash incentive for businesses.
- # Do not include video in employee questionnaire packets.
- # Use \$10 pre-incentive for employees (this was the most cost-effective plan, although the \$10/\$10 hybrid plan had slightly higher response rates).
- # Use first class stamps on return envelopes (further testing of the type of return envelope will be carried out in the first year of the *O*NET Data Collection Program*).

Exhibit A-4 presents estimated response rates combining all experimental conditions.

A.1.4 Overview of the O*NET Data Collection Program

The *O*NET Data Collection Program* will establish the foundation for continuous data collection to update and maintain the O*NET database, using collection methods designed to obtain high quality and current data. The primary method to update O*NET will be a general employer sample, that is, a survey of establishments and workers within those establishments. It will employ a two-stage, establishment-based survey design, including a statistical sample of businesses expected to employ workers in the specific occupations being surveyed and a sample of workers in the occupations within the sampled businesses. These sampled workers will be asked to complete the survey instruments.

Exhibit A-4. Employee Response Rates by Incentive Plan and Type of Return Envelope, No Video Included



O*NET uses four survey questionnaires: Skills, Knowledge (which includes Education and Training and Work Styles), Generalized Work Activities, and Work Context. These questionnaires will be used with sampled workers identified using the two-stage sample design. Each sampled worker will be randomly assigned only one of those four questionnaires. They will also be asked to provide basic demographic information and to complete a brief task inventory for their specific occupation. In addition, they will be offered the option of completing the questionnaire online. Data for a fifth domain, Abilities, will be provided by

trained analysts. Again, **Exhibit A-2** lists the questionnaires, the number of items, scales, and data sources.

For selected occupations, such as ones with small employment size, ones in which employees work in remote locations, and ones for which no employment data exist for use in the sample design, O*NET will use two other methods, either in addition to or in place of the primary method, that may offer effective and efficient approaches. The first of these methods involves sampling from member rosters of professional associations that include a significant portion of the occupation's workers in their membership. The second method will identify and use subject matter experts (SMEs) to complete the survey instruments. Each SME will be asked to complete the above mentioned four domain questionnaires, the demographic items, and the task inventory for the specific occupation being surveyed. More details regarding the use of SMEs and associations are provided in **Sections B.2.10** and **B.2.11** respectively.

The *O*NET Data Collection Program* has been designed to reduce the burden on businesses and employees in the following ways:

- # Due to O*NET's transition to the SOC, data will be collected on only 974 occupations, rather than on 1,122 O*NET 98 occupations.
- # Each incumbent will respond to only one of the four questionnaires.
- # The number and complexity of questionnaire items were reduced prior to conducting the pretest. Each questionnaire now requires only 25-30 minutes to complete.
- # The collection of demographic information from employers on non-responding employees, as used in the pretest, will be dropped.
- # Informational materials that will be sent to businesses and employees have been streamlined.
- # Procedures for sampling employees within a business have been simplified and automated.
- # Use of the web-based questionnaire will be encouraged by more prominent identification of this option in survey materials.

A.1.5 Summary of the Data Collection Procedure

The *O*NET Data Collection Program*'s primary method, the general employer sample, will involve the following broad steps, which are discussed in more detail in later sections.

1. Sample design and selection. The first stage of the sample, the sample of businesses, is based on the most currently available Occupational Employment Statistics (OES) data developed by the Bureau of Labor Statistics to identify industries in which workers in the occupation are expected to be found. The OES Survey data provide a distribution of employment in the occupation by Standard Industrial Code. The sample of establishments for an occupation will be drawn in proportion to each industry's share of employment in the occupation. The second stage, the sample of workers within the sampled businesses, will be selected randomly from among all workers in the establishment who work in the occupation, with the assistance of a "point-of-contact" in the business. Each sampled worker will be randomly assigned one of four survey instruments. **Sample design is discussed in detail in *Section B.1*.**
2. Solicitation and Data Collection. Following an extensive outreach effort, the survey staff will contact the sampled businesses to solicit their participation and to identify the specific "point-of-contact" to assist in the survey. This individual, possibly a human resources manager, will assist in drawing the sample of workers for the occupation. Survey materials will be mailed either to the point-of-contact for distribution or directly to the sampled worker at his or her work address. Each sampled worker will be asked to complete the survey during non-work time, using either the pencil/paper or the Internet version, and, for pencil/paper users, to return the response directly to the survey processing center. Survey operations will be conducted using an automated control system, and data entry will be hand-keyed. A nominal gift will be provided to the point-of-contact, and a non-cash incentive will be provided to the business. Monetary incentives will be given to all sampled workers. ***Sections B.2 and B.3.4* describe the survey procedures and outreach effort, respectively. *Section A.3* describes the automated tools to be used (control system and Internet). Incentives for respondents are described in *Section A.9*.**
3. Data Analysis and Publication. The data will be "cleaned" using standard edit procedures. The data will be summarized by computing the mean and standard error of the ratings for each descriptor, by occupation. Detailed analysis of non-response will be conducted. In addition, a variety of analyses are planned, focusing on reliability and validity of the ratings, demographic characteristics of the respondents, and others. The data will be released annually for public use through updates to the O*NET 3.0 database. Technical information such as sampling error measures will also be made available to users in hard copy and electronic reports. **Data analysis and publication are discussed more fully in *Section A.16*.**

4. Schedule. The data collection will proceed over the course of three years, each of which contains three collection periods. The schedule is presented in *Exhibit A-9* within *Section A.16*.

A.1.6 Statutory and Regulatory Information

Because O*NET is a relatively new system, it is not yet referenced specifically in statute or regulation. However, its predecessor, the *DOT*, is cited either directly or indirectly in a number of places as a source of occupational information in support of Federal programs. In addition, the Workforce Investment Act of 1998 contains a number of requirements and references to basic skills, skill trends, and other similar information, and has an extensive section on employment statistics. While O*NET is not mentioned specifically in the Act, it will be the foundation for carrying out the responsibilities related to these sections.

Section 309 of the Workforce Investment Act requires the Secretary of Labor to oversee the “development, maintenance, and continuous improvement of a nationwide employment statistics system” which shall include, among other components, “skill trends by occupation and industry.” Similar statewide employment statistics systems will be developed by the States. The Secretary’s first annual plan, prepared in response to Section 309, includes among its objectives the establishment of the O*NET data collection program and the incorporation of O*NET data into various workforce information products.¹⁰

The Workforce Investment Act contains numerous references to identifying the skill requirements of jobs; for example, Section 154 requires local determinations of the “skills and education that are necessary to obtain the employment opportunities” in the local area, and Section 134 requires providing “information on job skills necessary” to obtain jobs listed for the local area.

The Workforce Investment Act also contains references to identifying and assessing the skills of individuals; for example, Section 134 allows the provision of assessment services to identify “the skill levels and service needs of adults and dislocated workers.” Section 129 allows funds to be used to provide youth with an assessment “which shall include a review of basic skills, occupational skills, prior work experience, employability, interests, aptitudes (including interests and aptitudes for nontraditional jobs), supportive service needs, and developmental needs of such

¹⁰Quality Information...Informed Choice: New Directions for the Workforce Information System, Workforce Information Council, 1999.

participant....” In Section 136, the “attainment of basic skills and, as appropriate, work readiness or occupational skills” may be included as performance indicators for youth programs.

Section 508 of the Workforce Investment Act and Section 118 of the Perkins Act specify that persons with disabilities should have access to and use of the same information and data that persons without disabilities have. O*NET complies with this section by providing both low-vision and text-only versions of its online application that is used to access the data and information in O*NET. O*NET OnLine also provides links to several accommodation and disability resources on the Internet. Furthermore, the O*NET Consortium website, the Department of Labor’s Education and Training Administration O*NET website, and O*NET OnLine have all been *Bobby Approved*, indicating that the Center for Applied Special Technology has deemed them accessible to persons with disabilities.

The DOT, the predecessor to O*NET, is cited in the Code of Federal Regulation (CFR) in thirty places, shown in ***Exhibit A-5***. These citations generally are related to determining disability, administering of Department of Labor employment and training programs, or administering immigration, civil rights, and labor standards law. As the replacement for the DOT, O*NET is anticipated to serve as a source of occupational information in support of many of the programs referenced in these citations.

O*NET can also provide the framework or common language from which federal programs, including workforce investment and training programs of the Departments of Labor, Education, Defense, and Health and Human Services can develop their own guiding principles and regulations.

Copies of the sections of the Workforce Investment Act, the Perkins Act mentioned above, and the specific CFR citations are presented in ***Appendix B***.

A.1.7 Federal Register Notice

The Federal Register Notice announcing submittal of this request to OMB is presented in ***Appendix F***.

Exhibit A-5. Code of Federal Regulation Citations for the *Dictionary of Occupational Titles*

Title	Section	Regulatory Entity	Topic
20	220	Railroad Retirement Board	Determining Disability
	220.131		Vocational Considerations—Work which exists in the national economy
	220.132		Vocational Considerations—Physical exertion requirements
	220.134		Vocational Considerations—Medical-vocational guidelines
	220.187		Vocational Considerations—Medical-vocational guidelines
20	404 P	Social Security Administration	Determining Disability and Blindness
	404.1566		Work which exists in the national economy
	404.1567		Physical exertion requirements
	404.1569		Listing of Medical-Vocational Guidelines
20	416 Part I	Social Security Administration	Supplemental Security Income for Aged, Blind and Disabled
	416.966		Work which exists in the national economy
	416.969		Listing of Medical-Vocational Guidelines
20	627.240	Department of Labor	Job Training Partnership Act, On-the-job Training
20	651.10	Department of Labor	General Provisions Governing the Federal-State Employment Service, Definition of terms used in parts 651-658
20	653.103	Department of Labor	Migrant and Seasonal Farm Worker Job Applications
20	655, 656	Department of Labor	Temporary Employment of Aliens in the United States
	655.730		Labor Condition Applications and Requirements for Employers Using Non-immigrants on H-1B Visas in Specialty Occupations and as Fashion Models
	655.940		Attestations by Employers Using F-1 Students in Off-Campus Work
	656.21		Labor Certification Process for Permanent Employment of Aliens in the United States, Applications for labor certification for Schedule A occupations

Exhibit A-5 (continued)

Title	Section	Regulatory Entity	Topic
22	42.33	Department of Justice	Visas: Immigrants subject to Numerical Limitations. Diversity immigrants, Determinants of Work Experience
28	50.14	Department of Justice	Statement of Policy Regarding Equal Employment Opportunity, Guidelines on Employee Selection Procedures
29	553	Department of Labor, Wage and Hour Division	Application of Fair Labor Standards Act to Employees of State and Local Government
	553.30		Determination of exemptions for occasional or sporadic employment
	553.103		Determination of “same type of services” for exemption of volunteers
29	1607.15	Equal Employment Opportunity Commission	Documentation of impact and validity evidence—job titles and codes
34	600.2	Department of Education, Office of Post Secondary Education	Institutional Eligibility under the Higher Education Act of 1965 as amended, Definition of “recognized occupation” for which institution provides training
38	21.4802	Department of Veterans’ Affairs	Vocational Education and Rehabilitation, Service Members Occupational Conversion and Training Program, Definition of “related occupation” for purposes of employment of separating military personnel
41	60-3	Department of Labor, Office of Federal Contract Compliance Programs, Equal Employment Opportunity	Uniform Guidelines on Employee Selection Procedures, to assist employers, labor organizations, employment agencies and licensing and certification boards to comply with requirements of Federal law prohibiting employment practices which discriminate
	60-3.15		Documentation of impact and validity evidence
44	325.3	Federal Emergency Management Agency	Emergency Health and Medical Occupations, for planning emergency training and assignment of personnel

A.2 Uses/Products and Services Based on O*NET

O*NET, as the replacement for the DOT, serves as a viable and greatly enhanced resource for various critical federal functions. O*NET integrates a powerful relational database and a common language for occupation and skill descriptions into a value-added tool for workforce development professionals, businesses, and job seekers. By providing a database of information organized according to the O*NET Content Model, O*NET becomes an important tool for keeping up with today's rapidly changing world of work. O*NET OnLine, a web-based application allowing access to the O*NET 3.0 database, offers:

- # Detailed information on more than 900 occupations.
- # Descriptive information on standardized descriptors of skills, abilities, interests, knowledge, work values, education, training, work context, and work activities.
- # Standard Occupational Classification (SOC) coding.
- # Easy-to-use interface and search menus.
- # Skill-searchable occupational descriptions for more exact career targeting.
- # Links to accommodation information to use as a counseling tool for disabled workers.

When specific information from each user is combined with O*NET information, tailored solutions can be designed for multiple Human Resource (HR) applications. O*NET can help workforce professionals:

- # Forecast HR requirements more effectively.
- # Identify interest and experience requirements of different occupations to expand career options for job seekers.
- # Design relevant career curricula and occupational skills development programs.

Businesses and HR professionals can use O*NET to:

- # Expand the pool of quality candidates for open positions.
- # Develop job descriptions quickly and easily.
- # Define employee and/or job-specific success factors.

- # Align organizational development with workplace needs.

Job Seekers can use O*NET to:

- # Find out which occupations fit with their interests, skills, and experience.
- # Explore career profiles for growing occupations based on the latest available labor market data.
- # Maximize earning potential and job satisfaction.
- # Know what it takes to be successful in their field and in related occupations.
- # Identify skills that are transferable from one occupation to another.

O*NET can also be used by training facilitators, students, workforce researchers, career counselors, efficiency experts, rehabilitation counselors, software developers, displaced workers, business forecasters, industry analysts, educators, and organizational consultants.

The practical applications of occupational information found within O*NET are described below. O*NET's valuable role in career development initiatives and career information activities affects millions of youth and adults in this country. For example, incorporating O*NET into a system application designed to deliver career information offers the following advantages for specific audiences:

- # *A dislocated worker* who is looking for a new job can use O*NET in several different ways. First, the worker can look at O*NET information for his/her current job to identify what knowledge, skills, and abilities are required, enabling the worker to get a better idea of work-related strengths. The worker can then use O*NET to identify related occupations that require similar knowledge, skills, and abilities. Once the worker has identified new occupations of interest, the worker can then learn more about the characteristics and requirements of each of them using O*NET. The worker can identify what level of education and training is typically required for each occupation and can view the occupation's outlook.
- # *A rehabilitation counselor* can use O*NET to help clients identify their skills and then explore careers that require those skills. Clients can use O*NET to conduct an occupational search by skills. By having clients discover their marketable skills, the counselor can help them find an occupation that they find interesting and rewarding. Clients can also learn more about their current strengths and the skills they may want to acquire by exploring the skill definitions included in the Content Model section of O*NET.

- # An *employer* looking to expand and offer more employment opportunities within a community can also use O*NET. For example, the employer can use O*NET to better define the requirements of the occupational vacancies in the company. The employer could share that information with local workforce development programs to see if these agencies have clients with the required background. If not, arrangements could be made with local training program providers to develop training programs designed to meet the employer's needs. By using O*NET's common language, the employer, training providers, and job seekers can better understand the requirements of the employer's occupations.
- # A *front-line staff person* in a One-Stop Center can use O*NET to help identify the types of training programs in demand. By using O*NET's common language, the staff person could work with the employer to assess the type of skills the employer is looking for in new employees. Staff in the One-Stop Center could also identify Cross Functional Skills and Generalized Work Activities that cut across the various jobs the employer is looking to fill. By recognizing these general skills, workforce development professionals can then compare them to the qualifications of the local workforce and adapt training efforts to give workers the training they lack. The training program provider would also be able to prepare workers for several jobs within the employer's company, rather than just one particular occupation.
- # *Teachers* can use O*NET to develop career exploration activities for students. For example, students could use O*NET to match their interests and values to specific occupations. Faculty can develop curricula relevant to the world of work by using O*NET information about the knowledge, skills and abilities required for high growth occupations.
- # With O*NET, a *company* can greatly reduce the time it takes to perform effective job analysis. Using O*NET's hierarchical presentation of occupational information, the company can identify skills central to occupations within their organization — the important knowledge, skills, and abilities that workers must possess. The company can save time and money by identifying what work activities and core competencies are needed. Using O*NET's common language, a company can effectively communicate with the education and workforce development communities.

As these examples demonstrate, O*NET can serve as a key resource in addressing needs for many different audiences.

A.2.1 O*NET and the Education and Training Needs of the American Workforce

O*NET can assist training program operators, vocational and technical education program administrators, and planners in identifying skills and knowledges that are needed to meet changes in the world of work. They can then respond to these needs and offer training programs that will be important to the program participants and employers. Billions of federal, state and local dollars are used to support the occupational skill-specific training for millions of people, and O*NET supports more effective use of these funds.

The education, employment, and training systems in this country play a significant role in preparing for the future skill needs of the American workforce. Identifying common or transferable skills that are needed across occupations will become increasingly important as we adapt to changes brought on by a global, information-driven economy. Accomplishing this goal requires the ability to document the skill requirements of current and future occupations in the economy and to design curricula that are responsive to these needs so students can be prepared for future work opportunities. A slightly larger difference in response rates was detected for those receiving stamped (46%) versus business reply (43%) return envelopes ($p=0.04$).

Information compiled through the collection of O*NET data will benefit many programs, including the Occupational Outlook program of the Bureau of Labor Statistics. O*NET provides a common language for efforts to develop skill standards for specific occupations. Employers and economic development planners use a variety of occupational, educational, and labor market information for business planning decisions; again, O*NET serves as the means of linking the various sources into a single system.

A.2.2 O*NET for Career Exploration and Career Development

Career planning is a multi-phase process that individuals undertake throughout their lives. It begins in childhood during the elementary school years and lasts until a person retires from the workforce. During this process, individuals internally synthesize information about themselves and information about the world of work. O*NET includes the following integral types of occupational and worker information for career decision making: 1) Occupation Descriptions 2) Tasks 3) Abilities 4) Work Activities 5) Knowledges 6) Education and Training 7) Skills 8) Work Styles, and 9) Work Context.

Not only does O*NET serve as a direct resource for this important career information, but it will also serve as a source for the value-added products of commercial career information delivery systems and assessment developers. Career information delivery systems provide occupational and educational information for individuals in the process of career exploration and decision-making.

For example, some of the most widely used career information delivery systems, *Choices*, *intoCareers* (formerly the national Career Information System (CIS)), and ACT's *Discover*, have incorporated O*NET in reports and search functions. Some of the occupational information provided in *Choices*, *intoCareers*, and *Discover* comes directly from O*NET. *Choices* has customized the O*NET information to fit their particular customers. Users of *Choices* retrieve and search for occupational information based on O*NET topics and factors (e.g., Basic Skills, Generalized Work Activities, Interests, Physical Abilities). Similarly, *intoCareers* has integrated O*NET in its system and has incorporated many of the O*NET data descriptors into its descriptive occupational information. Finally, *Discover* used O*NET to create their world of work map which is the basis for many of their programs. These systems are used by several million people annually.

Numerous commercial aptitude and interest inventories/instruments used in career decision-making also rely on information in O*NET. O*NET serves primarily as a resource base for tools that utilize its information. In the remaining parts of this section, examples are provided about how O*NET can help individuals in various phases of career exploration and development.

While career information begins with knowledge of occupations young people see most often in their immediate surroundings, it is important that they also be familiar with a wider array of choices. This is particularly true for young people whose opportunities for direct experience may be limited because of family financial resources or due to geographic isolation — whether in a remote rural area or in an inner city. Career information can help break stereotypical notions that inappropriately limit young people. O*NET provides in-depth occupational information that can help open the minds of young students to a wide variety of career opportunities. Teachers can use O*NET as a resource to develop career awareness and exploration curriculum. The private and public sector can use the occupational information in O*NET to develop specialized tools to help students learn about the world of work and begin to discover the career opportunities that lay ahead for them in the future.

Millions of secondary school students need career information as they move from the occupational awareness phase of career development to the choices phase, where they make education decisions that affect their working adult lives. Secondary students begin to discover more information about themselves that is crucial to their future career paths. For example, they learn what their strengths are, what is important to them, and what they like to do. By using the career information in O*NET, these students can begin to determine how their abilities, strengths, and interests match up with different occupations in the world of work. They can begin to identify the education and training requirements of careers that they might want to pursue. Students will be able to obtain a more realistic view of what it takes to become a physician, lawyer, or computer programmer. O*NET will enable students to make better choices in school (e.g., what courses they will need to take, how long they will have to go to school), so they can be prepared for the career they wish to pursue. Further, O*NET can assist teachers in developing academic curricula that will help students meet the needs and skill levels of future jobs.

Of course, college students also need career information. College is a time for exploring a variety of opportunities. Students frequently enter college on one particular career path and then, based on experiences once they enter school, identify new opportunities that sound exciting and interesting. O*NET can help keep these students abreast of the constantly changing world of work, and the new skills and training required to pursue today's highly demanding jobs. O*NET can help inspire these students to examine the opportunities open to them, and help them make educational choices to prepare them to achieve their goals and meet the challenges that will lie ahead.

The need for career information does not end with the completion of formal schooling. There are many individuals who are in the middle of or nearing the end of their careers who also need information contained in O*NET. For example, members of the workforce who leave their jobs because of plant closings, business failures, and corporate downsizing may need not only new jobs, but new occupations with commensurate retraining. The information collected for O*NET will help these individuals identify new and growing opportunities in the market place. It can help them identify occupations that require similar skills to those they currently possess and help them to identify the training and education they will need to qualify for new and challenging occupations.

Individuals who become disabled during their adult years may also need information to help make potential career transitions using their skills. People with disabilities have a particular need for the career information that will be collected and contained in O*NET; that is,

information that accurately portrays the skill requirements of occupations. When this description of skills is in-depth, it supports accurate self-appraisal but does not erect artificial or inappropriate barriers to employment.

As some organizations offer early retirement incentives, middle-aged Americans seek new opportunities to use the talents they have developed during their years of work. Alternatively, workers retiring from one career may choose to pursue a new occupation in their retirement years. Older Americans often continue to work for pleasure or to augment fixed incomes after retiring from their primary job. This group needs the career information that will be collected for O*NET to be fully aware of available opportunities.

O*NET includes critical information that aids in career development. All types of people will be able to use O*NET and the value-added products developed using O*NET to make important career decisions, including the very young, those in the middle of their working lives, individuals with special needs, and those entering their retirement years. How people make decisions about the work they will pursue and the education and training they seek is affected by the information available to make career decisions.

A.2.3 O*NET as a Tool of The Workforce Development System

Human resource, education, business, government, and training professionals can use O*NET to stay ahead of a tight labor market and our rapidly evolving economy. Using O*NET's powerful, relational database, some have already created effective, timesaving systems for accessing the latest available labor market information and meeting workforce demands. Following are a few impressive O*NET success stories.

Employers

When Boeing prepared to close its aircraft manufacturing plant in Monrovia, California, managers and workers teamed with the California Employment Development Department to identify workers' skills and align those skills with available jobs. They used O*NET's skills survey tool to identify the workers' skills, and O*NET's customized reports to prepare a Skills Match report for workers. As a result of O*NET, nearly 400 Boeing workers found positions at other Boeing facilities *prior* to lay off. With information about their skills, and how those skills transfer to other occupations, workers were able to expand the scope of their job search.

State Agencies

Connecticut and Texas used O*NET to assess current and emerging job opportunities and the set of knowledge, skills, and abilities needed to obtain them. As a result of O*NET, Connecticut identified the 12 highest ranking skills needed to obtain jobs in the current top 25 occupations in the State. Texas identified the knowledge, skills, and abilities needed to obtain employment in 54 emerging or evolving occupations in 8 industries. Both states saved time and expense in identifying and targeting resources to address gaps. Employment and training communities tailored instruction and training to match occupations in demand. Both States improved the efficiency of gathering and presenting occupational data.

Information on High-Technology Occupations

An analysis of the multimedia and entertainment industries in California revealed a pressing need for a pool of talented and skilled 3-D computer artists and traditional animators. Using O*NET's survey data collection instruments and other materials, California's Employment Development Department conducted an industry study, with the endorsement of the California Skillsnet Consortium — comprised of businesses and educators. As a result of O*NET, human resource personnel modified local training and education initiatives to help fill the gap between industry needs and the skills possessed by local workers. Training programs defined the skills and requirements for the evolving occupations of 3-D computer artists and traditional animators. State and local leaders forged important partnerships with local businesses, educators, Workforce Investment Boards, community-based organizations, and other stakeholders to meet industry demands.

Connecting Education and Work

Several states use O*NET to tie their School-to-Work efforts with local occupational demands. For example:

- # New York developed **CareerZone**, an interactive, online system powered by O*NET for students, youth, and job seekers. Students can build resumes, generate cover letters, fill out job applications, learn more about entry-level jobs, and explore future career opportunities.
- # Minnesota created an innovative system using O*NET that documents students' skills and work styles and tests them with local businesses.

- # Texas uses O*NET to fuel its career exploration CD-ROM software, **OSCAR** (Occupation and Skill Computer-Assisted Researcher), developing links between O*NET data and skills taught in the classroom, and integrating industry skill standards and industry-based job analyses into OSCAR to meet the needs of the business community.

Workforce Investment Act

The Connecticut Department of Labor (CTDOL) used O*NET to respond to the Workforce Investment Act requirement that states assess: 1) current and future job opportunities in the state; 2) the skills necessary to obtain those jobs; and 3) the skills necessary to meet the economic development needs of the state. To meet these requirements, they collected labor market information and published an extensive report entitled *Choices Today—A High Performance Workforce Tomorrow*. Occupational skills information featured in the report was extracted from O*NET. *Choices Today* provides labor market information including:

- # economic and workforce trends
- # growing industries in Connecticut
- # occupations with the greatest demand
- # skills needed for these high demand jobs
- # *Six Industry Clusters*, cluster occupations, and needed skills
- # occupational supply and demand
- # current workforce investment activities.

O*NET was essential to the *Choices Today* report. Section Four of the report, which describes skills necessary for Connecticut's high demand occupations, depended on O*NET data and Content Model information, especially Skills and Skill Descriptions. Section Five of the report presented O*NET skills needed by clusters of industries that are a focus of economic development in the State.

Choices Today was specifically designed for those involved in Connecticut's One-Stop employment and training system. Stakeholders in this system include community leaders, policy-makers, planners, educators, counselors, service providers, and program operators at state and local levels.

By using O*NET, the CTDOL was able to save time and meet the planning needs of the State workforce development system. Identifying skill requirements was a major part of the *Choices Today* report. Having skills information available through O*NET in a simple, easy-to-access format allowed CTDOL staff to focus on collecting other data required by the Workforce Investment Act.

A.3 Uses of Information Technology

The *O*NET Data Collection Program* employs the latest in information technology (IT) systems and procedures in order to enhance the quality of the data collected and minimize the burden on both the responding businesses and job incumbents.

O*NET data collection had previously been implemented as a mail survey using hardcopy instrumentation. The *O*NET Data Collection Program* will continue to use hardcopy forms as the primary method of collecting the data. An Internet-based system has also been developed to allow job incumbents the option to respond to a questionnaire electronically. The inherent advantages of an electronic collection include improved data quality (through the incorporation of real-time edit checks) and more rapid receipt of the data (electronically-reported data eliminates the need for mail delivery time and post-collection data entry).

Many of the benefits inherent in the hardcopy form were replicated in the electronic questionnaire. Specifically, users are able to start the form and stop, prior to completion, multiple times without losing data. They can return to the partially completed questionnaire at any time during the survey period and continue at the point where they left off. A respondent may also review and edit previous answers if needed. In addition, an on-screen meter continually appears on the screen to keep the respondent informed as to how much of the total questionnaire is completed and remains. The programming effort involved in developing an on-line data collection application is a one-time expense whose product can be used for numerous cycles of data collection.

Advances in Web technologies and security, as well as the prevalence of access to web browsers by businesses and employees, have led to the conclusion that an Internet-based data collection is both feasible and practical. Growth in Internet use in the past few years has been enormous, and growth in online use is expected to continue. The pretest provided a threshold level for on-line use of O*NET questionnaires with almost 4% of responses completed online. Efforts are underway to identify and evaluate means of promoting and encouraging the use of the on-line questionnaire option for the full study data collection. The questionnaire cover will highlight the option of filling out the survey on the web.

An O*NET data collection web application has been developed to be used with the express goal of supporting and assisting with the *O*NET Data Collection Program* [<http://www.onet.rti.org/>]. The site is divided into two major sections, referred to as the *public* and *restricted access* sections. The restricted access section is further subdivided into two areas, the online questionnaires and the project management information.

Public Section: The goal of the public section is to support the business recruiting process. This section is accessible to the general public without any restrictions. This area of the site includes O*NET background information, endorsement letters, Frequently-Asked- Questions (FAQs), copies of each of the questionnaires and links to other O*NET related web sites. The purpose of the public section is to provide businesses and sampled workers with readily-accessible information regarding the data collection effort and uses of the data.

Restricted Access Section: This section contains sensitive information which should only be made available to certain populations (survey respondents, project managers, data entry staff). It is therefore a restricted access application, controlled by a user ID and password authentication scheme. The web server includes a Secure Socket Layer (SSL) certificate to allow encrypted transmission of all information over the Internet. This is the same technology used by electronic commerce web sites to secure credit card numbers. Also, no “cookies” are used. A “cookie” is a piece of data given to a user’s web browser, so that the browser will hand it back to the server (or web site) upon subsequent requests. The restricted-access section is further divided into two main subsections, the online questionnaire area, and the project management area.

Online Questionnaire Area: This area contains online versions of all of the survey instrumentation, providing sample members with an alternative method to paper and pencil. Only individuals who have been selected to participate in the survey can have access to this area of the site. Unique user IDs and passwords are automatically assigned to each job incumbent by a Central Office computer system at the time they are selected into the sample. The ID and password are provided to the sample member with other survey materials. The web site can confirm the validity of the ID and password and verify that a completed survey form has not already been received before allowing access to the Online Questionnaire Area.

Once entering this portion of the site, the respondent is:

- # informed that participation in the survey is voluntary
- # assured their survey responses will remain confidential

- # only able to see the questionnaire that they have been asked to complete
- # not aware that the other questionnaires exist
- # permitted to stop at any point and continue responding later
- # permitted to skip questions that they choose not to answer, and
- # permitted to go back and review/change previous responses.

On the last page of the survey respondents are asked to confirm that they have completed the questionnaire, at which time they exit from the questionnaire area of the site and are thanked for their participation. Their user ID is automatically deactivated at this time. Any further attempts to log in will not be possible, as the system recognizes this user as having completed the survey.

The database containing the survey data is not accessible by the Internet, but resides on a server inside the contractor's firewall. The web data collection application serves as the "go-between" to transmit data from the respondent's computer into the survey database. Only project staff, operating from inside the firewall, have access to the master survey database.

Project Management Area: This area contains project management reports and information. Access accounts for this area are created for those individuals responsible for the management of the *O*NET Data Collection Program*. This portion of the web site serves as an Intranet for the study, thus facilitating communication among project and DOL staff. Production reports, which are automatically posted nightly, include summaries of the progress associated with business recruiting, questionnaire shipment and receipt, and overall project status. Additional applications include a document library that allows designated data entry staff to make necessary changes and edits to various documents used in the data collection effort. This document library serves as a central repository for these documents, therefore eliminating ambiguity and confusion regarding the validity and currency of multiple versions.

A.4 Efforts to Identify Duplication

As recommended by the Advisory Panel, O*NET is designed to be comprehensive, providing reliable and valid information for a wide range of variables for all occupations in the U.S. economy.

To avoid duplication, some portions of the O*NET Content Model will be provided from existing data sources. Specifically, as discussed in **Section A.1**, the domain of Occupational Characteristics — including information on industries, job opportunities, and pay

— will be provided through linkages with existing labor market information databases. Information about occupational licensing and about related instructional programs will be provided from existing sources.

The exhaustive reviews of existing labor market and occupational information conducted by the Advisory Panel and the Department of Labor’s DOT Review staff, and subsequent research identified no other comprehensive, valid and reliable sources that could be used for any of the data items included in the proposed collection. The development of O*NET has also involved staff and advisors with many years of experience in the fields of labor market and occupational information who are familiar with existing data sources. In fact, many existing systems that currently provide detailed occupational information are actually using information based on O*NET 98 (now O*NET 3.0) or the *Dictionary of Occupational Titles*.

The few existing sources with similar measures are too limited to be used in O*NET. Some existing sources are valid and reliable, for example information from the Office of Personnel Management and the Department of Defense. These sources are not comprehensive, however, as they represent only jobs in federal civilian employment or in the military. There are also some private sources of job analysis information, which may be derived from valid job analysis methods, but they are not representative of the entire workforce. These sources are based on job analyses conducted on an “as needed” basis, rather than on a representative sample of employers and workers, and are thus limited in their coverage. As private sources, these data are not available to the general public. They also are not based on the “common language” O*NET provides.

A.5 Efforts to Minimize Burden on Small Businesses

All sizes of businesses will be represented in the *O*NET Data Collection Program* in proportion to their total employees in the workforce. **Exhibit A-6** shows the estimated numbers of establishments and employees in private industry by size of business. This tabulation excludes employees in government (20 million) and self-employed persons (9 million).

Establishments in the smaller size groups account for a small percentage of total employees. There is a large turnover among the smallest establishments, which sometimes creates operational difficulties and makes data collection less productive. The participation of small businesses in O*NET is necessary, however, since a significant proportion of the

Exhibit A-6. Number of Establishments and March 1998 Employment by Size of Establishment, Private Industry, United States

Number of Employees	Number of Establishments	Percent of Establishments	Number of Employees	Percent of Employees
0 - 4	4,298,821	58.9	6,643,089	6.5
5 - 9	1,270,474	17.4	8,394,786	8.2
10 - 19	814,204	11.2	10,968,798	10.7
20 - 49	561,231	7.7	16,992,116	16.5
50 - 99	193,315	2.6	13,317,500	12.9
100 - 249	112,734	1.5	16,946,459	16.5
250 - 499	28,996	0.4	9,855,812	9.6
500 - 999	11,009	0.2	7,491,354	7.3
1,000 +	5,774	0.1	12,182,772	11.8
Total	7,296,558	100.0	102,792,686	100.0

Source: U.S. Department of Labor, Bureau of Labor Statistics, Office of Employment and Unemployment Statistics, Division of Occupational and Administrative Statistics, ES-202 Program.

employment for some occupations occurs in small businesses. Therefore, all sizes of establishments will be sampled in the O*NET data collection program.

The burden on small businesses as a group is minimal since they will be sampled at a very low rate. Further, participation in the O*NET survey program does not place a significant burden on those small businesses selected in the samples. The major effort of participating in an O*NET survey is the task of preparing lists of employees in one or more occupations; this is a minimal effort for a small business. Analysis of data from the 1999 O*NET pretest indicated that the smaller businesses were more likely to cooperate with O*NET than the larger businesses.

A.6 Consequences if Information Collected Less Frequently

A.6.1 Consequences of No Data Collection

O*NET, and previously the *Dictionary of Occupational Titles*, is the most comprehensive standard source of detailed occupational information in the United States. No other similar comprehensive and reliable source is available.

If the data are not collected, O*NET users will have few options, none of which will meet their needs. They may continue to use the obsolete *DOT*, as many of them have done since the discontinuance of job analysis activity to update the *DOT*. Already these users are missing key information on changes in jobs which have taken place in the last decade and, given its age, may question the applicability of all of the data in the *DOT*. For example, consider the changes to the world of work in recent years resulting from the revolution in information technology. In addition to new occupations resulting from emerging technologies such as the Internet, most jobs in the workplace today have been significantly impacted by the use of the computer.

Over 10,000 of the 12,762 occupations in the *DOT* were last updated in 1977. Users who continue to rely on the *DOT* also will not have the new information items to be collected in O*NET or the advantages of its new data structure and standard classification. They will have to continue to struggle with how to relate the *DOT* to information on the labor market, as the *DOT* does not use a standard occupational classification. They also will continue to struggle with the unwieldy *DOT* data, which contains over 12,000 occupations and narrative information in a nonstandard format.

Users may also use the interim O*NET 3.0 database. As described in **Section A.1.1**, O*NET 3.0 uses existing information from the *DOT*, re-cast into the O*NET Content Model and updated based on analyst review. O*NET 3.0 is a major improvement over the *DOT*, as it is more current, in a standardized classification structure and format, and in an electronic form that is easy to use. However, since O*NET 3.0 utilizes source data derived from the *DOT*, it is based on dated information. Without the prospect of updates to the data provided by the *O*NET Data Collection Program*, users will be forced to continue to use dated information for the important decisions they must make.

Without the O*NET data collection, users might turn to various data sources which are neither comprehensive nor statistically based. For example, information from job openings

listed with the Employment Service might be used to identify employer hiring requirements. This information, while useful, represents only the jobs the Employment Service receives (i.e., it is not representative or statistically valid) and can provide only a few of the elements contained in O*NET.

The consequence of not conducting the *O*NET Data Collection Program* is that the millions of users who need O*NET to make the important life, business and policy decisions, as described in **Section A.2**, will be forced to make these decisions using obsolete information, incomplete information, and information of questionable validity and reliability. These users will not have benefit of practical results from the publicly funded research which has led to O*NET. Further, O*NET users, as well as the occupational information and the industrial/organizational (I/O) psychology fields, will not have the benefit of continuous improvement and further research in the measurement of worker characteristics and work requirements that O*NET will provide.

There are no legal obstacles to reducing respondent burden. Technical issues related to response burden are addressed in part B of this request.

A.6.2 Frequency of Data Collection

The *O*NET Data Collection Program* is designed as a continuing data collection to populate and refresh the database regularly in order to reflect changing labor market and skill requirements. The current plan is to produce a complete database in three years. During this time, a schedule will be developed for future data collection activities.

A.7 Special Circumstances

The study will be conducted in a manner which is entirely consistent with the guidelines in Title 5, Section 1320.6 of the Code of Federal Regulations. There are no special circumstances that might require deviation from the guidelines.

A.8 Consultation Outside the Agency

The following individuals have been consulted in the development of this submission. All have been asked to comment on the entire range of issues covered in the supporting statement; however, not all have reviewed the final clearance package.

Name	Organization	Telephone Number
Department of Labor, not in Employment and Training Administration		
Shail Butani	Bureau of Labor Statistics	202-691-6347
John Galvin	Bureau of Labor Statistics	202-691-6400
Michael Pilot	Bureau of Labor Statistics	202-691-5703
Alan Dorfman	Bureau of Labor Statistics	202-691-7378
Federal Government, Outside Department of Labor		
Patrick Cantwell	Bureau of the Census	301-457-8105
Sylvia Karman	Social Security Administration	410-965-7693
Nancy Kirkendall	Energy Information Administration	202-426-1088
David Pollock	Immigration and Naturalization Service	202-305-0081
Non-Government		
Michael Campion	Purdue University	765-494-5909
Steve Heeringa	University of Michigan	734-764-4424
Janet Wall	ACT Educational Technology Center	410-584-8000

As mentioned in **Section A.1.7**, *Federal Register Notice* initiating the 60-day period for public comment was published on August 7, 2000 (see **Appendix F**). The public comment period closed on October 6, 2000. Six individuals or organizations submitted a total of fifty-seven comments. These comments are summarized below, along with the response from the Department of Labor, Employment and Training Administration (ETA).

Data Collection Approach. Seven comments were received on the overall data collection approach. Two of these comments supported the plan to use associations in the data collection. One comment suggested that the use of household surveys should be investigated, although they recognized that cost might be prohibitive. The supporting statement has been modified to address the alternative types of surveys, including household surveys.

One comment addressed the use of job incumbents versus job analysts to provide O*NET data, noting that analysts are trained and experienced at assessing differences in skills and abilities. We note that there are a variety of potential rater sources for collecting occupational information. End-users place a premium on information gathered from individuals who “do the work,” i.e. job incumbents. I/O psychologists often point to the potential biases and errors likely to be found in ratings from particular sources.

Similar questions about appropriate respondents were raised with the O*NET pretest, which used job incumbents as the source of data for all domains. In response, a panel of experts

in industrial/organizational psychology was convened to review the appropriateness of using alternatives to job incumbents as the source of data for each of the O*NET domains. For the abilities domain, the panel did recommend that job analysts be used, based on consensus that the error likely present in job incumbent ratings outweighed the benefits of gathering information from individuals on the job. However, for the other domains, including the skills domain, there was no consensus to use a method other than the job incumbent survey. Thus the O*NET Team has determined to continue with collection of skills from job incumbents. As part of our continuous improvement efforts, we will review the quality of data collected, including potential rater source biases.

One comment noted that the experience level of the job incumbent, as well as other demographic characteristics, may have an effect on response. ETA understands this potential, which we believe is addressed by obtaining a representative sample of job incumbents for each occupation. One comment recommended that collection be conducted by the state employment statistics agencies. ETA does not consider this to be practical, as the survey is a national sample, and the difficulty of collection requires a well-controlled centralized survey operation. One comment raised the concern that the survey of incumbents might be inadequate as a means to identify emerging occupations. ETA plans to use other methods of information-gathering and research to identify new occupations that need to be added to O*NET and to the data collection process.

Role of the Point-of-Contact. Seven comments concerned the role of the point-of-contact (POC). These comments generally indicated the critical role of the POC in the survey, and the need for improving their rates of cooperation and reducing burden. ETA is well aware of the critical role of the POCs, and great emphasis is being placed on methods to improve their cooperation and reduce burden (see **Section B.3**). One of the comments questioned whether POCs might keep the cash incentives meant for job incumbents. The supporting statement has been clarified to indicate that the POCs are not made aware of the existence of the cash incentives.

Uses of O*NET. Six comments concerned the uses of O*NET. All of these comments were supportive of the need for and value of O*NET. One comment suggested that the description of uses in the draft supporting statement was overstated. This section has been re-written. One comment concerned whether O*NET would meet the needs of vocational rehabilitation users. This comment included recommendations for cooperative efforts between the Department of Labor and the rehabilitation community to ensure these needs are addressed.

The Department has an ongoing dialogue with representatives from the rehabilitation community and will further explore the feasibility of such cooperative efforts.

Response Bias. Four comments concerned possible response bias, specifically whether job incumbents will provide inflated ratings, whether job incumbents and subject matter experts have different response biases, and whether the web-based questionnaire has been validated as being equivalent to the paper-and-pencil questionnaire. Regarding the ratings from different types of respondents, this issue has been extensively reviewed over the course of O*NET's development, as discussed on the previous page. Regarding the web-based questionnaires, **Section A.3** has been expanded to describe the care that has been taken in designing the web questionnaire to be equivalent to the paper-and-pencil version in appearance and function, as well as in content. Also, the number of web-based responses expected is small. ETA will monitor this issue, however.

Sampling frame. Four comments addressed the employer sampling frame's coverage and quality. One comment concerned the use of the InfoUSA employer list instead of the ES-202 employer list from the Federal-State Covered Employment and Wages program. While ETA would prefer to use the ES-202 list, it is not available because of confidentiality restrictions. These comments also concerned whether the sampling frames covered the intended population, i.e., all workers, and whether workers such as the self-employed will be covered. The survey procedures have been modified to provide information concerning self-employment coverage.

Sample Size. Four comments questioned whether the sample size is adequate. The issue of sample size has been extensively studied in the development of the O*NET sample design. The O*NET Team maintains that the planned sample size will provide results that meet the standard which has been set for the O*NET ratings (see **Section B.1.1**). One comment concerned whether the sample size is large enough to capture the possible heterogeneity within an occupation. This comment addresses the issue of whether the O*NET occupations are defined in enough detail, that is, that differences may be greater within occupations than between occupations. The extensive analysis of O*NET data, as discussed in **Section A.16**, will provide insight on this issue, which will be used in future refinement of occupational classification.

Questionnaires; English proficiency. Three comments concerned how job incumbents with limited English proficiency will be able to respond. The O*NET Team will initiate a

project for translating the questionnaires into Spanish, although this effort will not be completed in time for initial data collection.

Questionnaire response time. Three comments concerned the estimated questionnaire completion times. One comment said the estimated completion times seemed reasonable, while two comments questioned the estimate of response time. The supporting statement indicates that response time estimates are based on timings conducted prior to the Pretest, and timings from the Internet responses during the Pretest.

Questionnaire difficulty. Three comments concerned the difficulty of completing the questionnaires. High response rates within questionnaires suggest that responding job incumbents did not experience difficulty with particular questions. However, the supporting statement has been modified to clarify the commitment to continuing work to improve questionnaire design, format, and ease of response, as well as to indicate the very high level of item response rates in the Pretest.

Nonresponse Error. Three comments concerned the potential for nonresponse error. Two of these comments questioned whether use of the web-based questionnaire would encourage higher response from job incumbents who have Internet access than from others. One comment concerned possible bias resulting from differences between job incumbents who elect to respond versus those who don't.

Privacy. Three comments concerned maintaining the privacy of job incumbent respondents. One of these comments questioned potential for violating privacy as demographic data on nonrespondents are collected from the POC. This procedure, which was used in the Pretest, has been dropped because of privacy concerns and low response rates. One comment concerned privacy of the web-based questionnaire; **Section A.3** has been modified to indicate that the system does not use "cookies." One comment concerned the effect of the disability questions in the background section of the questionnaire on response rates, due to privacy concerns. The O*NET Team recognizes this problem, but believes the need to analyze the representativeness of the respondents with respect to disability outweighs this concern. Furthermore, demographic data will be analyzed by occupation as a whole and no identifying information on firms or respondents will be carried in the database that provides composite ratings.

Miscellaneous Survey Design Comments. Three comments were received on various survey design topics. One comment concerned verification of whether the job incumbent is working in the occupation of interest. The supporting statement has been modified to clarify the procedures in this regard (see **Section A.16**). One comment questioned the use of different job incumbents for each questionnaire, although the comment noted that reasonable alternatives are not apparent due to response burden. The reason ETA chose to use four separate survey questionnaires was to reduce required response time and thus reduce burden and increase response rates. One comment suggested that, while the planned five-year cycle for refreshing the O*NET data is probably satisfactory, more frequent collection may be needed in some rapidly changing occupations. ETA agrees that the cycle for refreshing the database needs more examination, and the supporting statement has been modified accordingly.

Analysis and Dissemination Plan. Two comments addressed analysis and dissemination of O*NET data. One comment recommended conducting validity studies to examine the accuracy of job incumbent responses. The selected data collection approach is based on extensive research and testing. At the present time ETA is focusing on collecting data for nearly 1000 occupations to populate the O*NET database with current information. One comment requested that standard deviations for ratings be provided. Standard deviations, as well as standard errors, will be included in the database.

Assistance to Respondents. Two comments suggested the need for job incumbents to seek answers or clarification from the survey staff. The survey materials in **Appendix D** include toll-free telephone numbers and e-mail addresses for respondent use in asking questions.

Sample Design - Sex and ethnic representation. One comment concerned whether the sample design would result in respondents who are representative of the sex and ethnic composition of job incumbents by occupation. ETA's response is that the sample is designed to be representative of all workers, and, as indicated in the supporting statement, the results will be evaluated on this issue by comparing the gender and race/ethnicity of the respondents to that of all workers, by occupation, using data from the 2000 Census.

Response Rate. One comment suggested that the projected response rate for job incumbents may be somewhat high. The projected response rate is based on combining the best design elements from the pretest and on other improvements specifically designed to increase response.

Occupational Classification. One comment recommended a change in the occupational classification. This comment will be referred to the Standard Occupational Classification Policy Committee.

A.9 Payments or Gifts to Respondents

Our primary objectives in offering incentives to participants in the *O*NET Data Collection Program* are to improve the response rate and statistical accuracy of our data collection program. First and foremost, our intention is to attain a high response rate from both organizations and individuals participating in the data collection effort. The incentives proposed are described below:

- # **Incentive for Employer:** O*NET Business Incentive Tool Kit that includes inexpensive aids or guides for using O*NET in selected human resource management or other business applications.
- # **Incentive for Point-of-Contact (POC):** \$10 desk clock, or other inexpensive office supply item.
- # **Incentive for Respondent:** \$10 cash incentive included with questionnaire.

For employers, we propose an in-kind incentive that supports the objectives of the O*NET Program. The O*NET Business Incentive Tool Kit provides the businesses with guides to using O*NET and O*NET OnLine, as well as case scenarios to illustrate how O*NET can be helpful in human resource and other management applications. This serves to help employers understand the value of O*NET and will increase their cooperation and participation.

Incentives are appropriate when there is a gatekeeper who could prevent the respondent from ever receiving the questionnaire. In our data collection program, employers must agree to participate in identifying potential respondents (job incumbents) for the occupations we are seeking, and in distributing the questionnaire packets or in providing us with the work addresses of potential respondents so that questionnaires can be mailed directly to them. The business will be asked to designate a POC, usually someone in human resources management, to provide this assistance. Establishing a good rapport with the POC and gaining their personal cooperation are key to ensuring that questionnaire packets are delivered to the potential respondents. In addition to providing an in-kind incentive to the employer, we propose to provide a non-cash incentive (approximate value \$10) to the POC as a means of enhancing rapport and cooperation.

We will also offer a monetary incentive to job incumbents to ensure that, once distributed, a high percentage of those job incumbents respond by completing and returning the questionnaire.

Monetary incentives have the greatest potential impact when the respondent has to exert some special effort such as taking a test. The incentive serves to encourage respondents in a task requiring higher levels of involvement and commitment than the typical one-time face-to-face interview. While the O*NET questionnaires are not tests, the cognitive demands they place on respondents are quite similar, in that the respondents as workers in given occupations are asked to make assessments of the requirements of their job. Using various scales, workers must reflect on their current job in order to make these judgements. We view the monetary incentive as being instrumental in impressing upon the respondent the importance of this rating task. Respondents who perceive the rating task as important are less likely to make hasty assessments that lack adequate consideration, thus improving the reliability of the data.

In addition, the monetary incentive for respondents is important because we are encouraging them to complete the questionnaire on their own time rather than on the job. This serves to minimize the burden on employers and to improve the quality of the data since a nonresponse by workers who were "too busy" to complete the questionnaire on the job could produce a bias reflecting on job performance levels. Payment amounts were evaluated in the pretest to determine the optimal means to maximize the response rate (see *Exhibit A-4*). Based on that data, the \$10 cash incentive will be used for the full study data collection.

Finally, we see the monetary incentive as having the potential to at least partially offset its inherent cost through efficiencies created in the data collection process as a result of higher response rates. For the job incumbent respondent especially, while they are not viewed as a difficult-to-reach population in the usual sense, considerable effort and cost will be expended to identify and reach them through the sampling process. They represent a worker in a specific occupation in a specific establishment in a specific industry. The expense of reaching that particular respondent justifies the cost of a monetary payment to ensure a high rate of response.

A.10 Assurance of Confidentiality

All participants in this data collection effort will be assured through written materials that the confidentiality of their answers is in compliance with Section (a)(2) of the Wagner-

Peyser Act. In addition, very little personal information about the respondent is collected and there is no identifying information such as name or place of employment on the survey itself.

Research Triangle Institute (RTI), the contractor for the data collection task, has extensive experience in protecting and maintaining the confidentiality of respondent data collected from all types of surveys. RTI has drawn upon their experience in designing the data collection procedures incorporated in this program to assure confidentiality. In addition, all research involving human subjects is reviewed and approved by RTI's Institutional Review Board (Multiple Project Assurance number M-1496) prior to study implementation.

The collection of survey data is at the employee level at selected businesses and within recruited associations. The letter from the RTI project director, the letter from the Department of Labor, and the Participation Information Sheet for Employees that are sent to each selected employee (see **Appendix C**) contain essential program information that will enable the employee to make an informed decision regarding his/her voluntary participation in this national data collection effort. The commitment of project staff to maintain the confidentiality of information provided by the employee is clearly stated in those documents.

Employees are asked to complete the assigned questionnaire on their own time, not company time. This enables the employee to select a comfortable and private setting, if desired, in which to complete the questionnaire. The employee mails the completed questionnaire **directly** to RTI by using a reply envelope that is provided to them. The individual responses will be processed using a study ID number. All RTI O*NET project staff will be required to sign a Confidentiality Pledge that assures each respondent that the confidentiality of responses to the questionnaire will be maintained by RTI. No one except authorized project staff will have access to the completed instruments and data files. The completed and processed questionnaires will be stored in a secure document control area until federal authorization has been granted to destroy them. All computer files, including those associated with the control system, will be locked (password protected).

As previously discussed in **Section A.3**, the *O*NET Data Collection Program* will also utilize an Internet-based system to allow employee respondents to report data electronically. The respondents will have restricted access controlled by a user ID and password authentication scheme. The web server will include a Secure Socket Layer (SSL) certificate to allow encrypted transmission of all information over the Internet. This is the same technology used by electronic commerce web sites to secure credit card numbers. The database containing the survey data will

not be accessible by the Internet, but will reside on a server inside the RTI firewall. The web data collection application will serve as the “go-between” to transmit data from the respondent’s computer into the survey database. Only project staff, operating from inside the firewall, will have access to the master survey database.

A.11 Questions of a Sensitive Nature

The only question which might be considered of a sensitive nature is the item in the Background section asking respondents to identify whether they consider themselves to have a disability, using the disability questions developed for the 2000 Census of Population. However, completion rates obtained during the O*NET pretest indicate that the vast majority of participants (97%) elected to complete the question.

The sampling strategy used by the O*NET collection randomly selects participants at the individual level. The disability question, along with the demographic questions, will provide descriptive information on the sample of respondents. In addition, when the data are available, they may allow for some broad comparisons with the characteristics of the general population working in the occupation using information from the 2000 Census.

Respondents to the O*NET survey will be informed that responding to all questions, which includes disability status and the other demographic characteristics, is voluntary. The information will be protected from disclosure of the characteristics of an identifiable respondent by separating individuals’ identity from their response data, as will be described in **Section B.2.6**.

The O*NET pretest also included a set of questions used to gather from employers the basic demographic characteristics of non-responding employees. The questions covered employees’ job title, job tenure, age, gender, ethnicity, and race. The pretest collection was not successful in obtaining a significant amount of this information from employers. The overall response rate was 29%, with item response rates ranging from 9% to 28%. For this reason, along with potential concerns over employee privacy issues, these questions will not be included in the current O*NET collection procedure.

A.12.a Estimates of Annualized Hour Burden

The burden calculation is based on 300 occupations to be surveyed in year one of data collection. Hour burden is at the business/POC level as well as at the employee level. The

breakdown of burden hours by specific event is displayed in **Exhibit A-7**. These estimates of annualized hour burden for the businesses were calculated based on the pretest (see **Appendix E**). During the pretest, we found the telephone calls for the screening, recruiting, and sampling activities took much less time than previously estimated in the OMB pretest clearance package. The estimate of the burden associated with completing questionnaires is based on timings conducted prior to the pretest as well as timing data collected for the questionnaires completed via the web during the pretest.

A.12.b Estimates of Annualized Cost Burden to Respondents

The cost burden to employee respondents is \$207,270, based on ½ hour of time for 25,200 respondents at an hourly rate of \$16.45. The rate is based on 1998 Bureau of Labor Statistics (BLS) National Compensation Survey, inflated to 2000 rates using the Employment Cost Index.

Given that the POC often will be a Human Resource manager, we can derive an estimate of cost by using the 1998 mean hourly wage of \$25.10 for Personnel, Training, and Labor Relations Managers, published by the BLS. Again, the BLS Employment Cost Index is used to inflate the 1998 wage to the 2000 mean hourly wage of \$26.27. The estimated cost burden for the first year of data collection is \$281,220, based on 10,705 Total Business Burden Hours (from **Exhibit A-7**) at \$26.27 per hour.

The estimated total cost burden for employees and businesses together is \$488,490.

A.13 Annual Reporting Burden Cost

There are no respondent costs for capital or start-up, or operation, maintenance and purchase of services. There is no cost to employers, POCs, or sampled workers other than the time it takes them to comply with the survey request. The POC will more than likely participate during his/her work day. **Exhibit A-7** indicates that individual POC participation in the *O*NET Data Collection Program* will take about 1½ hours of time. The business verification information may be supplied by the POC or may be provided by someone else at the company.

The estimate of burden for this is 5 minutes per business, making the total burden per business/POC 1 hour and 35 minutes.

**Exhibit A-7. Estimate of Annualized Hour Burden for the O*NET Data Collection Program
(Based on 300 occupations being surveyed)**

Activities	Number of Respondents	Number of Responses per Respondent	Average Burden Hours per Response	Total Burden Hours
Verification Calls to Business	12,000	1	1/12	1,000
Initial Recruitment Call to Business	7,920	1	1/4	1,980
POC Creates Occupation Lists for Sampling	6,180	1	1/2	3,090
Call to POC to Sample Workers	6,180	1	1/6	1,030
POC Distributes Questionnaire Packets	6,180	1	1/3	2,060
Follow-up Calls to POC	6,180	3	1/12	1,545
Total Business Burden Hours				10,705
Questionnaires Completed ¹	25,200	1	1/2	12,600
Total Burden Hours				23,305

¹ Includes incumbents, Subject Matter Experts (SMEs), and association members.

A.14 Estimates of Annualized Cost to Government

The estimated cost to the government for the *O*NET Data Collection Program* (i.e., sampling, data collection, and data analysis) is \$5.8 million in the first year, \$5.8 million in the second year, and \$5.9 million in the third year of the data collection period. This is a new project; therefore, estimated operational costs (such as personnel, travel, supplies, overhead) for responsible members of the National O*NET Consortium were compiled in order to derive the estimated cost figures.

A.15 Reasons for Program Changes Reported in A.13 and A.14

This is a new project and will count as an increase of 23,305 hours, per *Exhibit A-7* (Program Change) towards Employment and Training Administration's (ETA's) Information Collection Budget.

A.16 Time Schedule, Publication, and Analysis Plans

The major components of the *O*NET Data Collection Program* include sampling, recruiting and training, data collection, and analysis. *Exhibit A-8* provides the expected schedule. The schedule assumes OMB clearance is received on or before January 22, 2001. A change from that assumption would require a recalculation of the data collection schedule. Data collection will be carried out using overlapping collection periods, with each period covering approximately one-third of the occupations to be included in each year's survey.

In this section, we provide our approach to data cleaning and the types of analysis we expect to perform. The results of these various analyses will suggest additional analyses. Most of these analyses will be conducted after the data files are closed upon completion of each data collection cycle. At the end of this section we list analyses that will be conducted annually.

A.16.1 Analyses Conducted at End of Each Data Collection Cycle

Data Cleaning

Paper questionnaires will be manually edited so that completely blank questionnaires are dropped, responses to items that should have been skipped will be blanked out, multiple responses will be blanked out, and codes indicating missing data, multiple responses, and

Exhibit A-8. Data Collection Schedule

ACTIVITY DESCRIPTION	SCHEDULE
Sample 300 Occupations for Year 1	November 15, 2000 - January 29, 2001
OMB Clearance Granted	January 22, 2001
Period 1 Data Collection	March 5 - August 6, 2001
Period 2 Data Collection	May 29 - October 29, 2001
UPDATE DATABASE	First Quarter 2002
Period 3 Data Collection	August 27, 2001 - February 18, 2002
Sample about 330 Occupational Units for Year 2	September 11 - November 7, 2001
Period 4 Data Collection	November 26, 2001 - April 15, 2002
Period 5 Data Collection	February 24 - July 29, 2002
Period 6 Data Collection	May 28 - October 28, 2002
UPDATE DATABASE	First Quarter 2003
Sample about 344 Occupational Units for Year 3	June 10 - August 7, 2002
Period 7 Data Collection	August 26, 2002 - January 27, 2003
Period 8 Data Collection	November 25, 2002 - May 26, 2003
Period 9 Data Collection	March 24 - August 25, 2003
UPDATE DATABASE	First Quarter 2004

legitimate skips will be inserted. Codes for legitimate skips and missing responses also will be inserted in the records for questionnaires obtained through the web.

Identification and Evaluation of Anomalous Cases

Analyses of the ratings are predicated on the assumption that the raters were qualified, willing and able to engage in the rating tasks. Potentially unusable ratings will be identified by flagging those who indicated that there were no tasks that were important to their job or who failed to respond to task information questions. Potentially unusable ratings for a particular questionnaire also will be identified by flagging those raters missing more than fifty-percent of his/her data and by flagging those with patterns of ratings that showed little or no discrimination among descriptors. Data from these raters then will be subjected to a rational review by a panel of three judges, who will make independent judgements regarding the usability of the record

according to a process and criteria developed by the O*NET Consortium. Responses judged to be invalid will be excluded from the analysis file.

Compute Sampling Statistics

Basic sampling weights will be applied to the data in order to make inferences to the population of incumbents for each occupation. These weights will be computed as the inverses of the overall selection frequencies, and the selection probabilities for each selected establishment and each individual participant. The analysis weights for the eligible sample units will be adjusted to compensate for unit non-response, for both establishments and employees. Weighting class adjustments will be made to compensate for the missing establishments and participants. The first stage adjustments will be defined by industry codes (general sample) and occupation (target samples); the second stage adjustments will be based upon occupations, or groups of occupations. Weights also will be adjusted to correct for duplication between the general and target frames (dual frame samples).

Sampling errors will be computed. The analysis weights used in the sampling error computations, as noted above, will have been adjusted for non-response and will be consistent with the complex sampling design.

Non-respondent Analyses

Frequency of non-response will be calculated by occupation and by questionnaire to identify O*NET domains or items that respondents refused to complete. In addition, as in the pretest data analysis, businesses that do not respond to the O*NET instruments will be analyzed to ascertain whether certain characteristics of an organization, such as size, location, and industry, or the organizational position held by the POC, predict willingness to provide O*NET data. Non-response analyses will use logistic regression analysis with respondent/non-respondent as the dependent variable and business characteristic variables as the predictors. Should such features be identified, this information could greatly inform subsequent data collection efforts by suggesting alternative ways of contacting or interacting with the POCs at those firms. If such attempts are successful, the result should be increased response rates in subsequent data collections.

Calculation of Descriptor Values and Reliability

For each occupation, the means and standard deviations of the ratings for each descriptor will be calculated. Standard errors will be calculated along with the 95% confidence interval around the mean. Following standard O*NET methods¹¹, descriptors with a standard error of at most 0.75 (a confidence interval half-width of 1.5 or less) will be considered reliable estimates of the rating. Occupations will be considered to be reliably represented if at least 95 percent of the descriptor standard errors are estimated at 0.75 or less.

A.16.2 Analyses Conducted Annually

Generalizability Study

Generalizability theory provides a mechanism for evaluating the representativeness of observed ratings by allowing researchers to isolate the components of variability in a set of data. Important potential sources of variation in this data collection will include the occupation, the rater, and the industry. The outcome of the analyses will be a better understanding of the sources of error in the ratings. Interventions can then be considered to reduce variation from the undesirable sources.

Confirmatory Factor Analyses of Domains

Confirmatory factor analyses provide an empirical demonstration of the plausibility of the theoretical structure believed to underlie the O*NET content model. However, some domains in the content model were not defined by an expected empirical relationship among descriptors; rather, they were defined to maintain comparability with theoretically-derived taxonomies or to represent a number of discrete descriptors. Therefore, the confirmatory modeling of the latent structure of the O*NET variable domains will be targeted to the domains for which we would expect covariation among descriptors.

¹¹Peterson, N.G., Mumford, M.D., Borman, W.C., Jeanneret, P.R., Fleishman, E.A., Levin, K.Y. (September, 1997). *O*NET Final Technical Report*. Salt Lake City, UT: Utah Department of Employment Security.

Create Occupation Database

The O*NET Database will be updated once per year. Each annual update will include data for those occupations collected and analyzed during that year. (Thus, an annual database update may include occupations from two, three, or four data collection periods, depending on the number of prior periods for which analysis was completed that year.) For each occupation collected, the newly calculated means data will *replace* existing data in the database. If elements in the questionnaires have been changed since the previous database update, an analysis will be done to define the impact on the existing database. As necessary, the database will be restructured and existing data will be converted to accommodate new/changed questionnaire elements.

The O*NET Database is designated with a version number that will be incremented with each annual update, e.g., from O*NET 3.0 to O*NET 4.0. The database is developed and administered using the Oracle database management system. Once the Oracle database is updated, it is used to generate the database for public release as a series of flat text files. It is accessible to the public via the Web site of the National Center for O*NET Development at www.onetcenter.org.

A.17 - Display of Expiration Date

The expiration date will be displayed on the cover of the survey questionnaires.

A.18 - Exceptions to Certification Statement

There are no exceptions.